

Positive families for the Schur weak- $\aleph$ question

FA/Banach run packet for arXiv:1412.1748

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Source question

The source paper is S. Gabrielyan, J. Kakol, W. Kubis and W. Marciszewski, *Networks for the weak topology of Banach and Frechet spaces*, arXiv:1412.1748. After proving that $\ell_1(\Gamma)$ is an $\aleph$ -space in the weak topology if and only if $|\Gamma| \leq \mathfrak{c}$, the authors ask:

Let E be a Banach space with the Schur property and satisfy (i)–(iii) of Proposition 3.4. Is E an $\aleph$ -space in the weak topology?

This packet does not settle the question in full. It records two positive families which are stable and useful test cases: weakly compactly generated Schur spaces, and Banach spaces isomorphic to subspaces of $\ell_1(\Gamma)$ for $|\Gamma| \leq \mathfrak{c}$.

Idea of the proof

The source already proves the hard nonseparable model case $\ell_1(\Gamma)$, $|\Gamma| \leq \mathfrak{c}$. The first observation is that the weak topology on a Banach subspace is exactly the subspace topology inherited from the ambient weak topology, because Hahn–Banach extends every functional on the subspace. Since $\aleph$ -spaces are hereditary to subspaces, the whole subspace family follows immediately.

The WCG family is different. If E has the Schur property, then weakly compact subsets of E are norm compact. A weakly compact generating set therefore has separable norm-closed span, so a WCG Schur space is separable. For a separable Schur space, a countable norm base is a countable k -network for the weak topology: weakly compact sets are norm compact and weakly open sets are norm open.

Lemma 1. *Let Y be a closed Banach subspace of a Banach space X . Then the weak topology $\sigma(Y, Y')$ is the topology inherited by Y from $(X, \sigma(X, X'))$.*

Proof. The inherited topology is generated by the restrictions $x'|_Y$, $x' \in X'$. It is therefore no stronger than $\sigma(Y, Y')$. Conversely, by the Hahn–Banach theorem, every $y' \in Y'$ is the restriction of some $x' \in X'$. Hence the two initial topologies are generated by the same linear functionals on Y . $\square$

Lemma 2. *Every subspace of an $\aleph$ -space is an $\aleph$ -space.*

Proof. Let $\mathcal{N} = \bigcup_n \mathcal{N}_n$ be a σ -locally finite k -network in X , and let $Y \subset X$. The family $\{N \cap Y : N \in \mathcal{N}\}$ is still σ -locally finite in Y . If $K \subset U$ with K compact in Y and U open in Y , write $U = V \cap Y$ with V open in X . Then K is compact in X , so there are $N_1, \dots, N_m \in \mathcal{N}$ such that

$$K \subset \bigcup_{j=1}^m N_j \subset V.$$

Intersecting with Y gives the required finite k -network refinement in Y . $\square$

Theorem 1. *Let Γ be a set with $|\Gamma| \leq \mathfrak{c}$. If a Banach space E is isomorphic to a closed subspace of $\ell_1(\Gamma)$, then $(E, \sigma(E, E'))$ is an $\aleph$ -space.*

Proof. By Theorem 1.2 of the source paper, $\ell_1(\Gamma)$ endowed with its weak topology is an $\aleph$ -space whenever $|\Gamma| \leq \mathfrak{c}$. If $F \subset \ell_1(\Gamma)$ is a closed subspace, Lemma 1 identifies $(F, \sigma(F, F'))$ with a subspace of $(\ell_1(\Gamma), \sigma(\ell_1(\Gamma), \ell_1(\Gamma)'))$. Lemma 2 gives that F is weakly an $\aleph$ -space. Finally, a Banach-space isomorphism $E \rightarrow F$ is a homeomorphism for the weak topologies, since its adjoint identifies the corresponding dual functionals. $\square$

Lemma 3. *Every separable Banach space with the Schur property is weakly $\aleph_0$.*

Proof. Let $\mathcal{B}$ be a countable base for the norm topology of E . We claim that $\mathcal{B}$ is a k -network for the weak topology. Let $K \subset U$, where K is weakly compact and U is weakly open. By the Eberlein–Smulian theorem and the Schur property, every sequence in K has a norm-convergent subsequence; hence K is norm compact. Also U is norm open. A finite subfamily of the norm base covers K and is contained in U . Thus $\mathcal{B}$ is a countable k -network for $(E, \sigma(E, E'))$. $\square$

Corollary 1. *Every weakly compactly generated Banach space with the Schur property is weakly $\aleph_0$. In particular, Question 4.9 has an affirmative answer for WCG Schur spaces satisfying the source paper’s conditions (i)–(iii).*

Proof. Let K be a weakly compact subset whose closed linear span is E . By the Schur property and Eberlein–Smulian, K is norm compact, hence norm separable. The closed linear span of a separable set is separable, so E is separable. Lemma 3 applies. $\square$

Limitations and obstruction

The result above is a genuine positive subcase but not a full solution of Question 4.9. The source proof for $\ell_1(\Gamma)$ uses a special annulus localization lemma: near every point of a thin norm annulus one can find a weak neighborhood whose intersection with that annulus has small norm diameter. This follows from the coordinate structure of $\ell_1(\Gamma)$. The Schur property alone only controls weakly convergent sequences and weakly compact sets; it does not by itself give such local norm control on arbitrary bounded annuli. Thus the quotient/general Schur-space case remains open in this packet.

Novelty check

Local run indexes were searched for arXiv:1412.1748, the source title, “Schur property”, “weakly aleph”, “countable Tychonoff”, and “ $\ell_1(\Gamma)$ weak topology”. Web searches for the exact Schur question and for later occurrences of “Banach space with the Schur property” and “weakly aleph” found the source paper but no exact later answer to Question 4.9. A later paper by Ferrando–Gabrielyan–Kakol titled *Functional Characterizations of Countable Tychonoff Spaces* has adjacent metadata about function-space characterizations, but the public abstract and local corpus did not expose an exact answer to this Schur-space question.

References

References

- [1] S. Gabrielyan, J. Kakol, W. Kubis and W. Marciszewski, *Networks for the weak topology of Banach and Frechet spaces*, arXiv:1412.1748.
- [2] M. Fabian, P. Habala, P. Hajek, V. Montesinos, J. Pelant and V. Zizler, *Functional Analysis and Infinite-Dimensional Geometry*, CMS Books in Mathematics, Springer, 2001.